

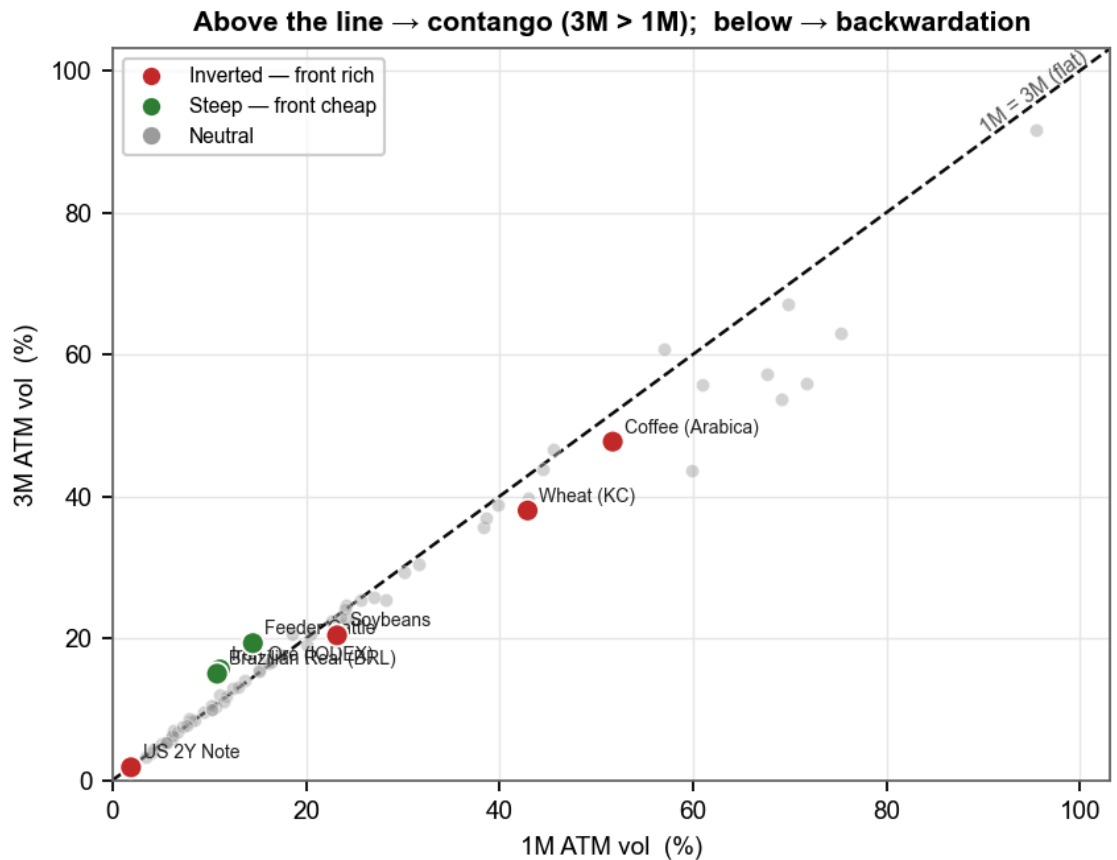
Vol Term Structure

Fri 24th July 2026 | XP Inc. Global Macro

Seven curves stand out today. Four are unusually **inverted**, near-term volatility bid above the deferred: **Soybeans, Wheat (KC), Coffee (Arabica)** and **US 2Y Note**. Three are unusually **steep**, the front trading cheap against the back of the curve: **Iron Ore (IODEX), Brazilian Real (BRL)** and **Feeder Cattle**. They are highlighted purely as markets where the curve currently looks stretched, and may be worth a closer look.

How to read this. Slope = $3M - 1M$ ATM implied vol; flagged when more than 1.5σ from its own one-year norm — **steep** = contango (front cheap vs the back), **inverted** = backwardation (near-term vol bid). Full methodology on the closing page.

CURVE SHAPE — 1M VS 3M



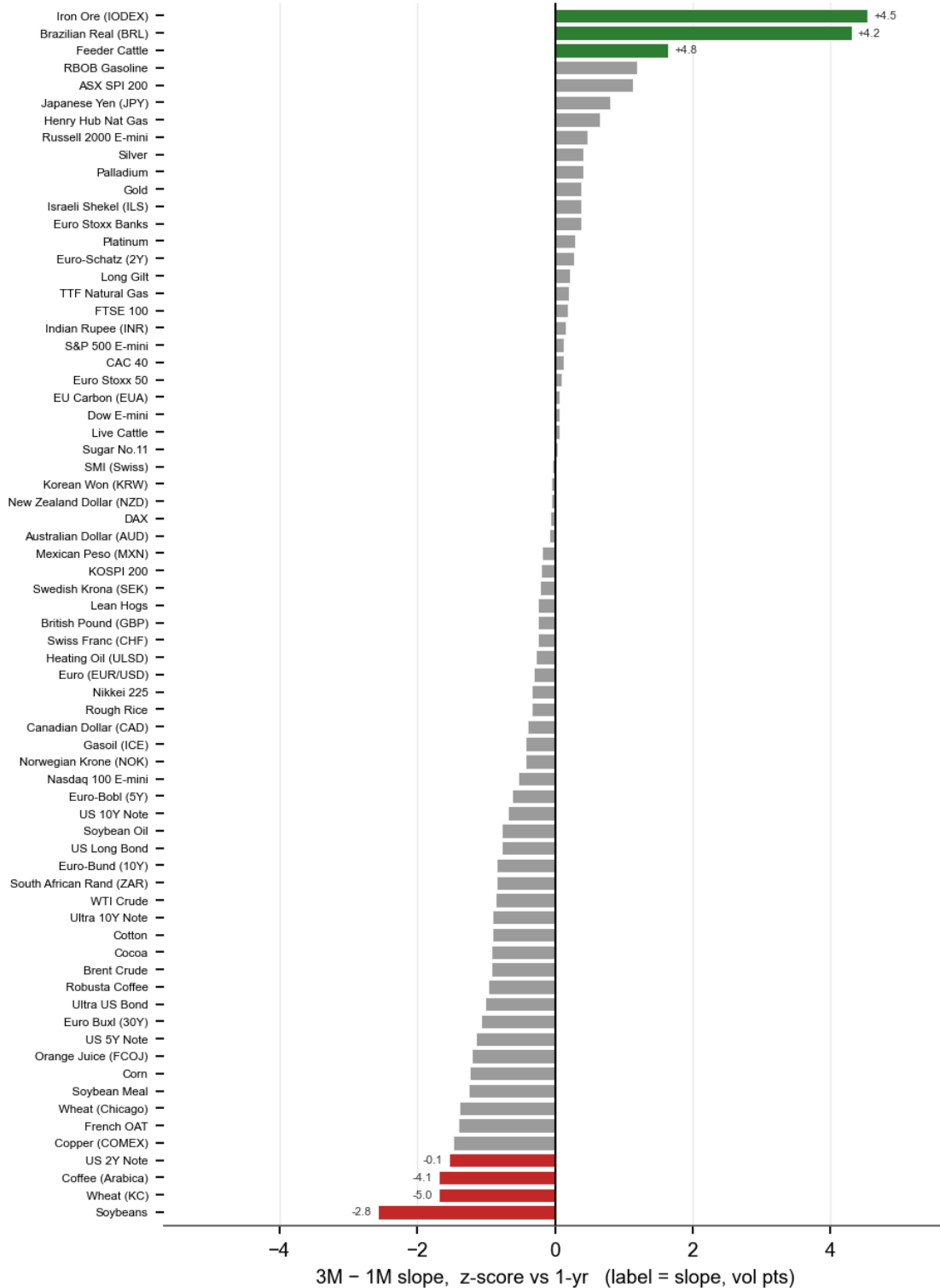
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MOST STRETCHED — RANKED BY 3M-1M SLOPE Z-SCORE

All markets ranked by spread z-score — flagged in colour

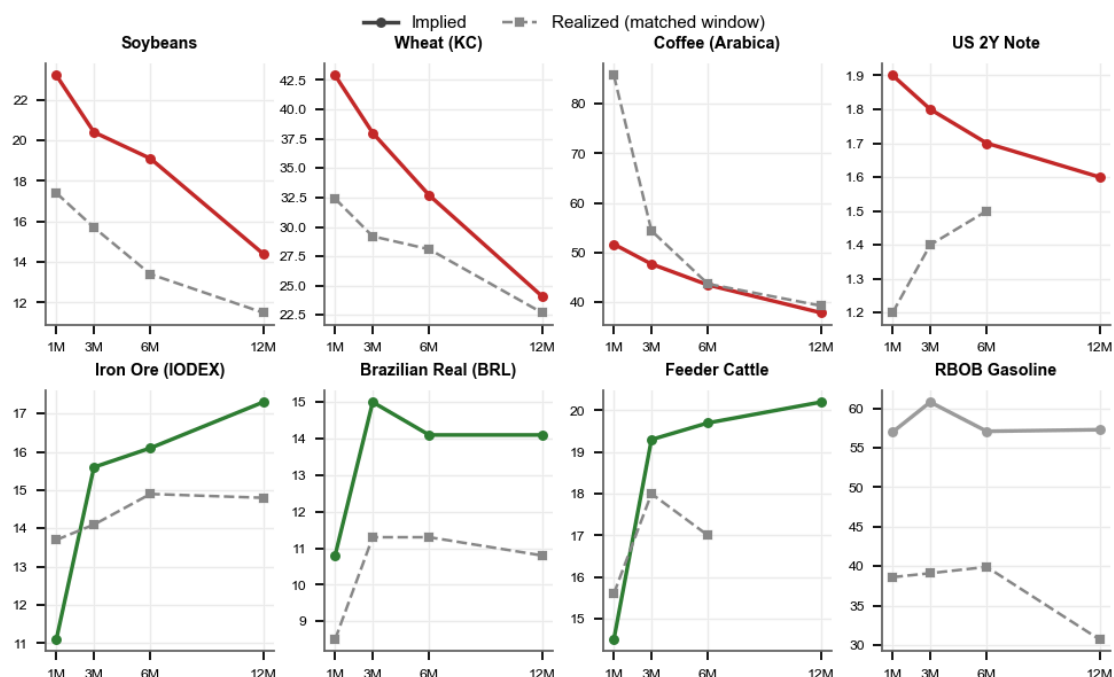


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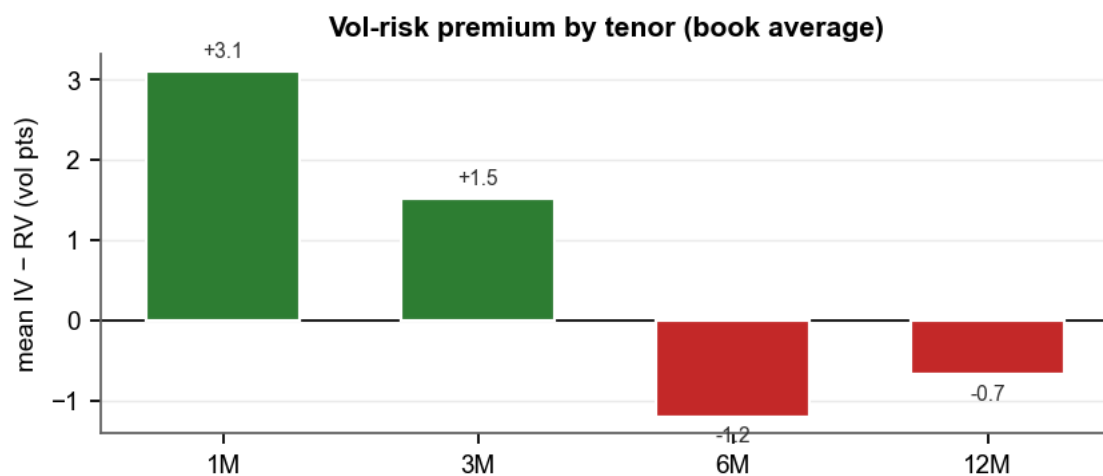
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CURVE SHAPES — IMPLIED (SOLID) VS REALIZED (DASHED)



VOL-RISK PREMIUM BY TENOR



FLAGGED OPPORTUNITIES

Market	Instrument	Signal	1M (±1σ)	3M (±1σ)	6M (±1σ)	12M (±1σ)	Slope	z (1y)	%ile
US 2Y Note · Bonds	TUA Comdty	Inverted — front rich ▼	1.9 (±0.122)	1.8 (±0.113)	1.7 (±0.112)	1.6 (±0.104)	-0.1	-1.54	4
Wheat (KC) † · Agriculture	KWA Comdty	Inverted — front rich ▼	42.9 (±20.14)	38.0 (±17.82)	32.7 (±15.36)	24.1 (±11.33)	-5.0	-1.69	5
Coffee (Arabica) † · Softs	KCA Comdty	Inverted — front rich ▼	51.7 (±10.23)	47.7 (±9.42)	43.5 (±8.61)	37.9 (±7.50)	-4.1	-1.69	7
Soybeans · Agriculture	S A Comdty	Inverted — front rich ▼	23.2 (±18.32)	20.4 (±16.09)	19.1 (±15.11)	14.4 (±11.35)	-2.8	-2.58	1
Feeder Cattle · Softs	FCA Comdty	Steep — front cheap ▲	14.5 (±3.117)	19.3 (±4.149)	19.7 (±4.229)	20.2 (±4.347)	+4.8	+1.65	97
Brazilian Real (BRL) · FX	BRA Curncy	Steep — front cheap ▲	10.8 (±0.134)	15.0 (±0.186)	14.1 (±0.174)	14.1 (±0.175)	+4.2	+4.31	100
Iron Ore (IODEX) † · Metals	SCOA Comdty	Steep — front cheap ▲	11.1 (±0.69)	15.6 (±0.96)	16.1 (±0.99)	17.3 (±1.07)	+4.5	+4.54	100

† our curve prints 3+ vols away from the vendor's 1M — Wheat (KC) +6.4 · Coffee (Arabica) +5.3 · Iron Ore (IODEX) -5.5. The table shows our numbers, taken from listed settlements.

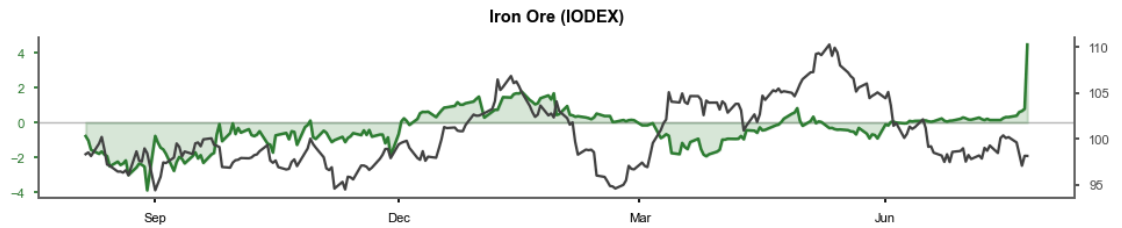
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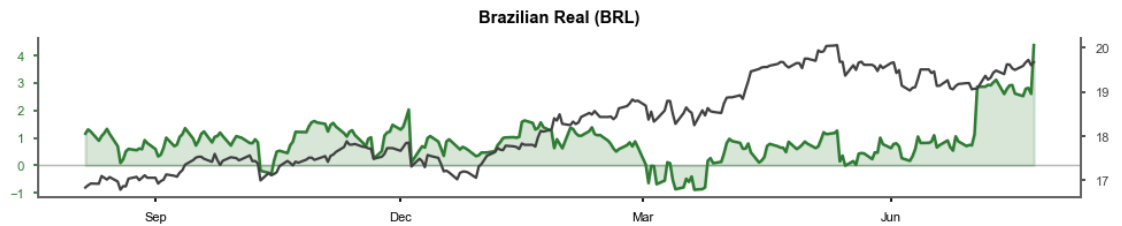
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FLAGGED — 3M-1M SLOPE VS UNDERLYING (1 YEAR)

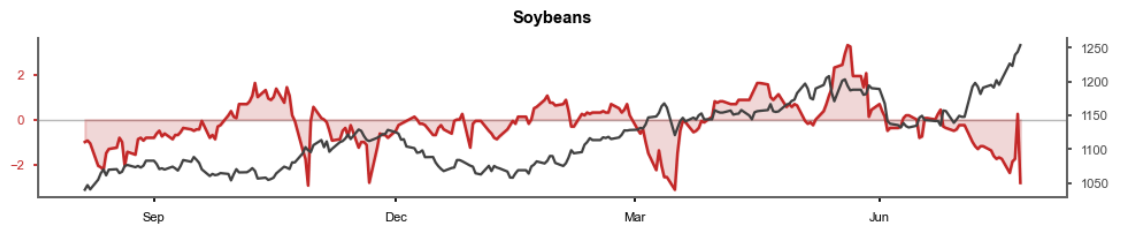
shaded / coloured: 3M - 1M ATM slope (vol pts, left axis; below zero = inverted) · grey: underlying price (right axis)



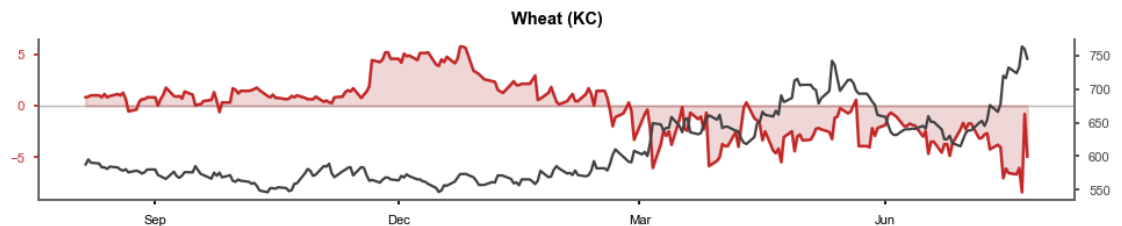
The curve has never been steeper over the past year, with the 3M-1M slope at **+4.5** vol points (1M **11.1** vs 3M **15.6**) — the **100th** percentile. The steepening has come from the front, with the 1M leg falling **-6.7** points over the past month against **-2.3** for 3M.



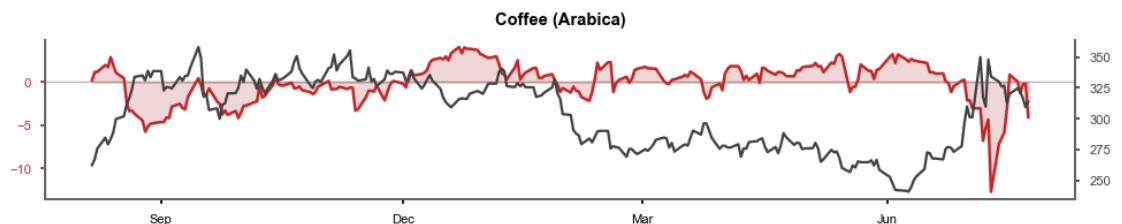
At **+4.2** vol points (1M **10.8** vs 3M **15.0**), the 3M-1M slope registers its **100th** percentile — the most steeply upward-sloping reading of the past year. Over the past month the 1M leg has drifted **-0.6** points while 3M has added **+2.7**, with the underlying up **+3%** over the same stretch.



Inversion has pushed the 3M-1M slope to **-2.8** vol points (1M **23.2** vs 3M **20.4**), a **1st** percentile reading last seen in **mid March**. This time the near-term bid stands out, with the 1M leg up **+5.8** points over the past month versus **+3.3** for 3M as the underlying has rallied **+8%**.



The 3M-1M slope is now **-5.0** vol points (1M **42.9** vs 3M **38.0**) — the **5th** percentile of the past year, and a degree of inversion last seen in **late April** when the underlying had moved **+7%** in a month. The current episode is bigger in every dimension: the 1M leg is up **+15.5** points over the past month against **+15.4** for 3M, with the underlying **+18%** over the same stretch.

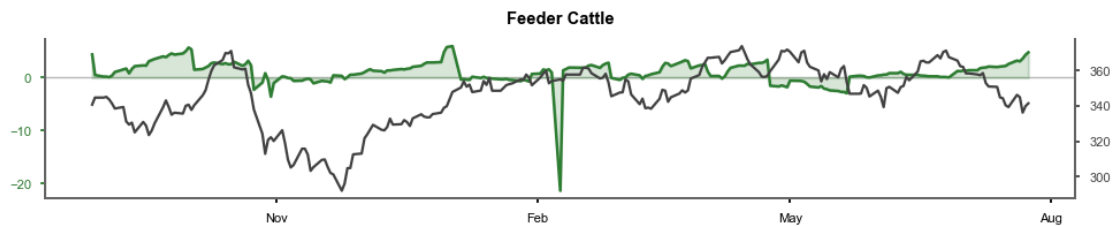


Sitting at **-4.1** vol points (1M **51.7** vs 3M **47.7**), the slope marks its **7th** percentile of the past year, a level of inversion last reached in **late September 2025** — though back then the underlying had moved **-4%** in a month. This time the whole curve has been dragged higher into a rally, with 1M up **+13.8** points over the past month, 3M up **+10.9**, and the underlying **+14%**.

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A steep **+4.8** vol points separates the tenors (1M **14.5** vs 3M **19.3**), putting the slope at the **97th** percentile — a shape last seen in **early January**, when the underlying had moved **+7%** in a month. The setup differs this time: the 1M leg is barely changed at **+0.1** points over the past month while 3M has risen **+4.8**, with the underlying down **-8%** over the same stretch.



Even at very low absolute levels, the curve is mildly inverted, with the 3M-1M slope at **-0.1** vol points (1M **1.9** vs 3M **1.8**) — a **4th** percentile reading last seen in **late March**. Moves have been modest, with the 1M leg up **+0.3** points over the past month against **+0.1** for 3M.

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DEFINITIONS & METHODOLOGY

The curve — ATM implied vol at constant **1M / 3M / 6M / 12M** maturities (listed option surface; FX: OTC tenor vols).

z-score & percentile — how stretched today's slope is against its own trailing year; $|z| \geq 1.5$ flags a curve, the percentile ranks it (100 = steepest of the year).

VRP by tenor — each tenor's implied minus the matched realized window (1M↔21d, 3M↔63d, 6M↔126d, 12M↔252d): where on the curve options are richest versus delivered movement.

Sources & timing. ATM implied vols at constant **1M / 3M / 6M / 12M** maturities are **our own construction** for 44 of the 70 markets: the same settlement-built curve behind the Volatility Report (ATM call/put settlements of every listed expiry, Black-76-inverted, fitted per product), read at each maturity and emitted only where the listed expiries genuinely reach it — longer tenors fall back to the vendor surface where they don't, as do FX (OTC tenor vols) and equity cash indices. † marks markets where our 1M and the vendor's disagree by 3+ vols. The realized comparisons behind VRP-by-tenor are our own calculation from settlement prices over the matched windows (21 / 63 / 126 / 252 days). Figures are as of the **Fri 24th July 2026** close, pulled 27 Jul 2026, 07:58.

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Slope — **3M – 1M** in vol points: positive = contango (the calm norm, the front cheap versus the back); negative = backwardation (near-term vol bid, typically stress).

(±1σ) brackets — each tenor's 1-day one-standard-deviation move in the contract's own price units (vol + $\sqrt{252} \times$ price), to the decimals it trades in.

Coverage — STIR (money-market) futures are excluded: their option vol is quoted in rate terms, a different convention from the price-based measures used here.

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